An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

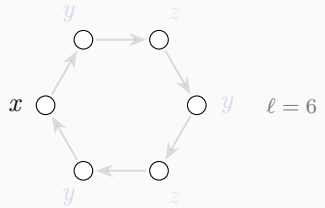
Yuya Uezato Yoshiki Nakamura

CyberAgent, Inc. Chiba University

STOC 2027 submission (author video for the PC)

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$\ell = 6$

$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

☛ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

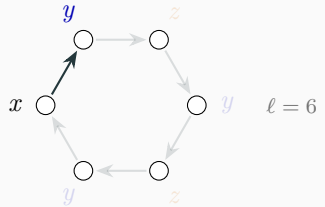
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- 辺を 1 本ずつ出し、式の対応する部分と一緒に読む
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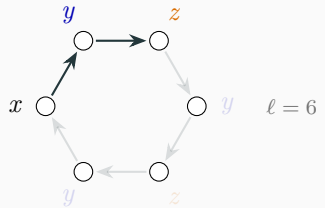
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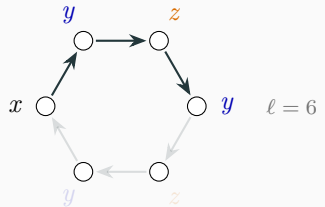
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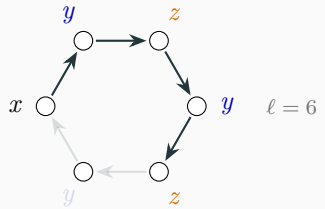
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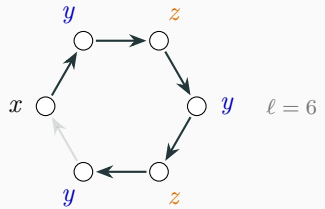
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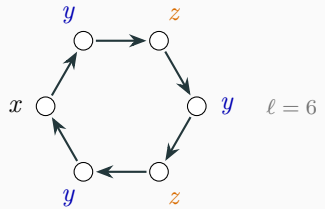
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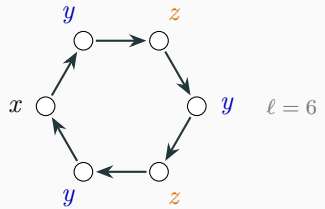
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FO^k model checking: $O(n^k)$ is 'optimal' for $k \geq 4$; the exception is FO^3

We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under HyperClique Hypothesis	[BS21, ICALP] [NU27, PODS]
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CoR [Tar41, JSL] Tarski's calculus of relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z.\ \cdots \wedge \cdots$, we use DNF translation.
$\rightsquigarrow$ $2^{O(\lvert\varphi\rvert)}$ compositions are used [Nak22].

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メモ：

- 合成 = ブール行列積
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Are $2^{O(|\varphi|)}$ compositions unavoidable?

DNF translation is used, for example:

$$\exists z.\ \big(P_1(x,z) \vee Q_1(z,y)\big) \wedge \cdots \wedge \big(P_k(x,z) \vee Q_k(z,y)\big)$$

$$\rightsquigarrow_{\text{(DNF)}}\ \exists z.\ \bigvee_{A \subseteq [k]} \Big(\bigwedge_{i \in A} P_i(x,z) \wedge \bigwedge_{i \notin A} Q_i(z,y) \Big)$$

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(In this translation, we use 2^k compositions, since the number of subsets $A \subseteq [k]$ is 2^k.)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS])

Can we prove that there is no polynomial-size translation from FO^3 to CoR?

Partial answer [Nak22, JLAMP]: yes for **existential** CoR [Nak23, LICS] ($\equiv$ existential FO^3), via preservation under extensions, which fails for full CoR.

スライド概要：この指数個の合成は必要か？ 未解決だった（$\Sigma_1\mathrm{CoR}$ のみ既知）。

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Outline

1. Our results

2. Technical overview

3. Conclusion

Our results

The answer is **YES!**

Thm. (This paper)

There is no polynomial-size translation from FO^3 *to* CoR.
More precisely, there are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$, binary — before:
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Our results

The answer is **YES!**

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There is no polynomial-size translation from FO^3 *to* CoR*.*
More precisely, there are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

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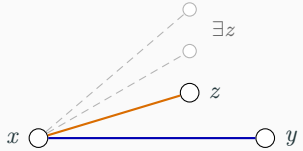
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The hard formula G_k

$$G_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k} \left[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z)) \right] \qquad (|G_k| = O(k))$$



we can pick z to satisfy: for every $i \in [k]$,
the x-y and x-z edges
have labels P_i and Q_i, in either order.

CoR: 2^k compositions

$$G_k \equiv \bigcup_{A \subseteq [k]} \left[\left(\bigcap_{i \in A} P_i \cap \bigcap_{i \notin A} Q_i \right) \cap \left(\left(\bigcap_{i \in A} Q_i \cap \bigcap_{i \notin A} P_i \right) ; \mathbf{1} \right) \right]$$

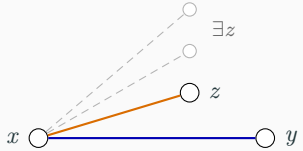
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Outline

1. Our results

2. Technical overview

3. Conclusion

Proof outline (at high-level)

Goal Lemma Every CoR term equivalent to G_k needs $2^{\Omega(k)}$ compositions.

We construct structures $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ such that

- G_k distinguishes all of $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$:

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G_k	T	F	F	$\cdots$	F	2^k	$2^k/2$

- **Key Lemma** If a CoR term t distinguishes d structures among $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$, then t needs $\geq d/2$ compositions.

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$	$\mathcal{M}_4$	$\mathcal{M}_5$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp. $\geq$
t	T	F	T	F	T	T	$\cdots$	T	2	1
t'	F	T	T	T	T	F	$\cdots$	F	4	2

☛ if $t \equiv G_k$, then t needs $\geq 2^k/2$ compositions.

TASK How can we construct such $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$?

スライド概要：G_k が $\mathcal{M}_0$ で成り立ち、$\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ で成り立たない構造を作る。d 個を $\mathcal{M}_0$ から見分ける項は、合成を $d/2$ 個以上必要とする。

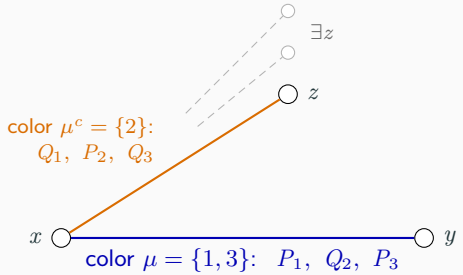
メモ：

- 表の T/F は、各 $\mathcal{M}_i$ に対応する点対での値（論文の test pair）
- 以降：構造を色付き完全グラフとして見る → $\mathcal{M}_0$（two-witness property、color-invariant）→ $\mathcal{M}_\lambda$（hub での flip）→ 合成が新たに見分けるのは高々 2 個である理由

Structures as colored complete graphs

View structures as **complete graphs**, colored with 2^k colors $\mu \subseteq [k]$: $P_i \iff i \in \mu$.

Q_i is used as the complement of P_i (i.e., $i \notin \mu$): exactly one of P_i, Q_i holds on each edge.



for every $i \in [3]$,
the x-y and x-z edges
have labels P_i and Q_i, in either order

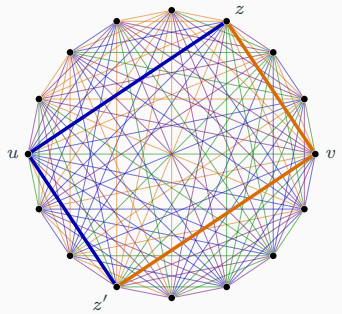
☛ $G_k(x, y)$ holds iff some x-z edge has the color μ^c, where μ is the color of the x-y edge.

スライド概要：構造を色付き完全グラフとして見る。G_k は「補の色の x-z 辺がある」と読める。

メモ：

- 各辺で P_i と Q_i のちょうど一方が成り立つ
- 例：$k = 3$、$\{1, 3\}$ と $\{2\}$ は互いに補

The randomly colored structure $\mathcal{M}_0$



$\mathcal{M}_0$: randomly colored, $O(k\,4^k)$ vertices
[JMT95, Algebra Univers.]

Two-witness property For all $u \neq v$ and all colors $\boldsymbol{\mu}, \boldsymbol{\nu}$, there are at least two **witnesses** z such that the u-z and z-v edges have colors $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$, respectively.

Every CoR term t is color-invariant on $\mathcal{M}_0$: $\mathcal{M}_0 \models t(u,v) \iff \mathcal{M}_0 \models t(u',v')$ whenever the u-v and u'-v' edges have the same color.

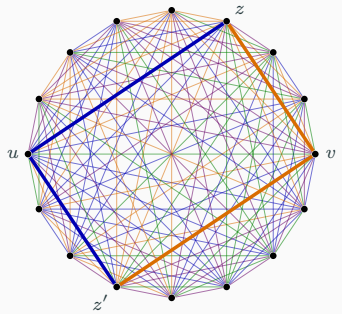
Proof by induction on t: the case, $t = t_1 \,;\, t_2$, follows from the (two-)witness property.

スライド概要：$\mathcal{M}_0$ はランダム彩色で、two-witness property をもつ。だからどの項も color-invariant（真偽は辺の色だけで決まる）。

メモ：

- 証明は項の帰納法；非自明なのは合成 $t_1 \,;\, t_2$ だけで、witness z の存在が色だけで決まる
- 「2 つ以上」が効くのは $\mathcal{M}_\lambda$ 側（hub を避ける witness が残る）

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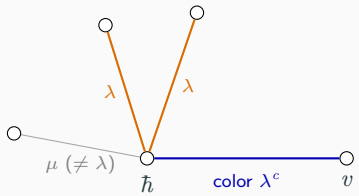
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$\mathcal{M}_\lambda$: flipping at the hub $\hbar$

Fix an arbitrary vertex $\hbar$ of $\mathcal{M}_0$, the **hub**.

structure $\mathcal{M}_0$



the hub $\hbar$ has λ-edges

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For v with the $\hbar$-v edge of color λ^c: $\mathcal{M}_0 \models G_k(\hbar, v)$ but $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$.
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$\rightsquigarrow$ Each composition newly distinguishes at most two of them (next slide).

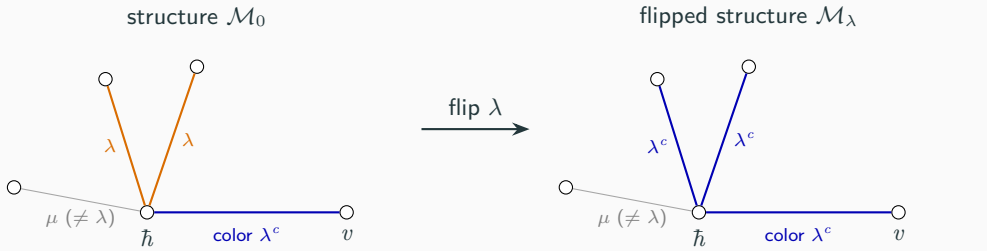
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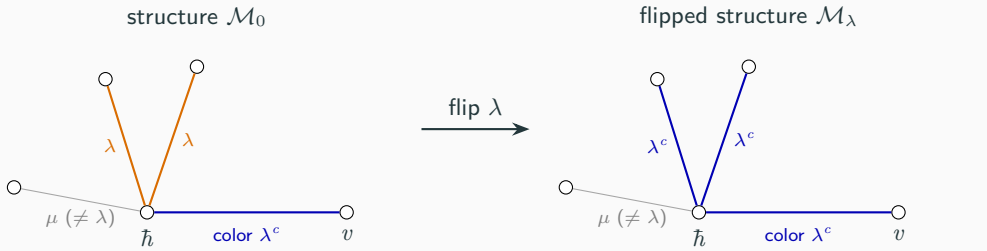
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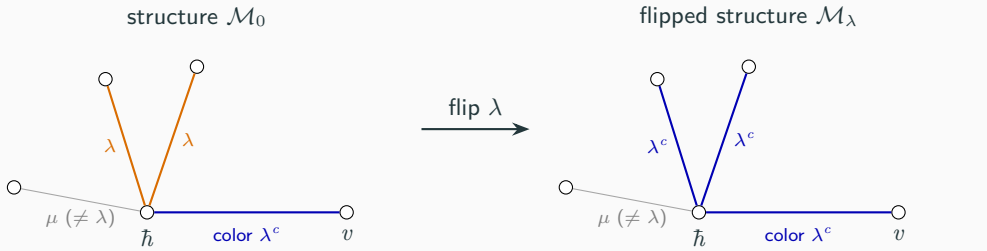
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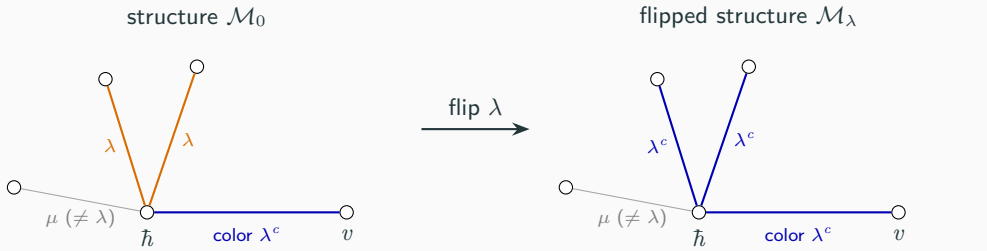
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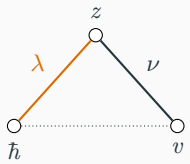
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Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$

$(X\,;\,Y)(u,v)$ holds iff there is a witness z; flipping λ changes only edges at the hub $\hbar$.
(in the panels, X holds on the λ-edges and Y on the ν-edges, $\nu \neq \lambda$)

(a) $\mathcal{M}_0$, at $(\hbar, v)$



witness z

At the hub (a, b): every witness uses a hub edge, so it can be lost
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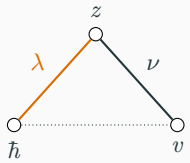
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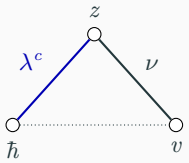
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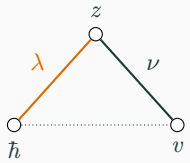
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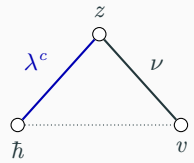
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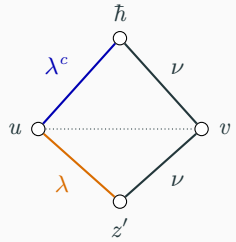
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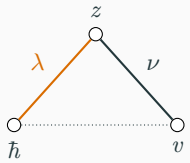
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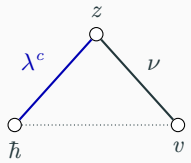
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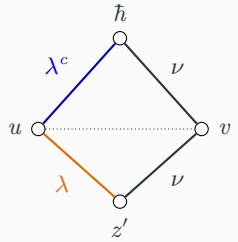
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Outline

1. Our results

2. Technical overview

3. Conclusion

Summary

Result Tarski's translation from FO^3 to CoR is optimal in the worst case:

- some FO^3 formulas of size $O(k)$ need $2^{\Omega(k)}$ compositions in every equivalent CoR term, even with complement, converse, and shared subterms;
- the same technique: FO^3 vs. FO^2, matrix query languages, CoR with closure.

Technique 2^k flips of one structure $\mathcal{M}_0$; each composition newly distinguishes ≤ 2 of them.

? Open questions

- a superpolynomial gap over a **fixed** finite signature (e.g., one binary relation)?
- the domain size from which the blowup begins?
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Thank you for your attention!

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- 未解決：固定シグネチャ、開始の領域サイズ、正確な合成数

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References i

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